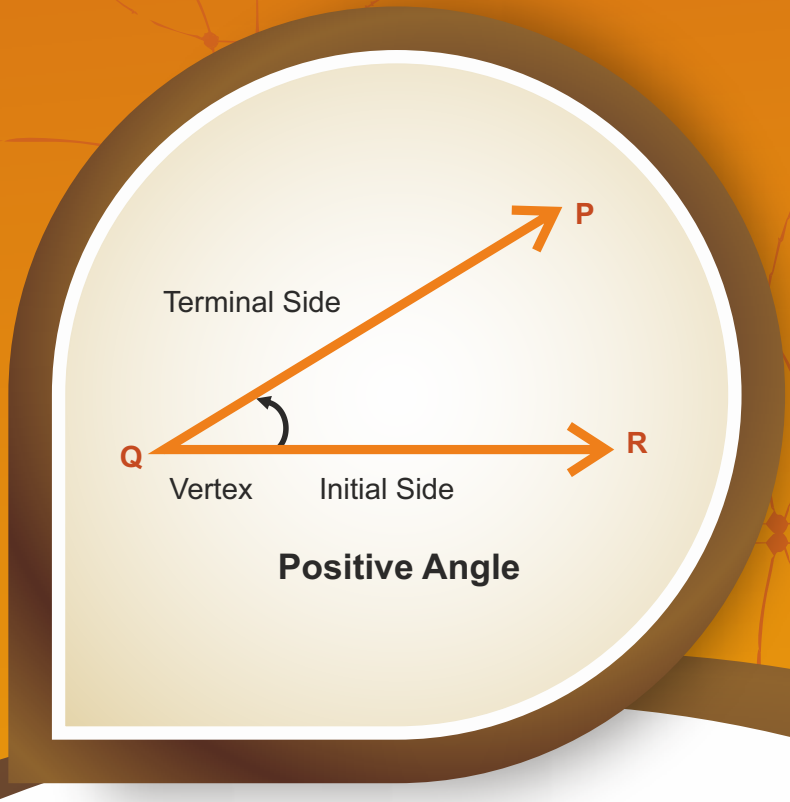
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Compound Angles

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is -

(A) 0 (B) 2 (C) 1 (D) 3

TR0001

2. $2(\sin^6\theta + \cos^6\theta) - 3(\sin^4\theta + \cos^4\theta) + 1$ is equal to -

(A) 2 (B) 0 (C) 4 (D) 6

TR0002

3. If $x = y\cos\frac{2\pi}{3} = z\cos\frac{4\pi}{3}$, then $xy + yz + zx =$

(A) –1 (B) 0 (C) 1 (D) 2

TR0003

4. The value of $\sin10^\circ + \sin20^\circ + \sin30^\circ + ... + \sin 360^\circ$ is equal to -

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) 2

TR0004

5. If $\cos(\alpha + \beta) + \sin(\alpha - \beta) = 0$ and $2010\tan\beta + 1 = 0$, then $\tan\alpha$ is equal to

(A) 1 (B) –1 (C) 2010 (D) $\frac{1}{2010}$

TR0005

6. If $\cos x + \cos y + \cos\alpha = 0$ and $\sin x + \sin y + \sin\alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$

(A) $\sin\alpha$ (B) $\cos\alpha$ (C) $\cot\alpha$ (D) $2\sin\alpha$

TR0006

7. $\frac{\sin(A-C)+2\sin A+\sin(A+C)}{\sin(B-C)+2\sin B+\sin(B+C)}$ is equal to -

(A) $\tan A$ (B) $\frac{\sin A}{\sin B}$ (C) $\frac{\cos A}{\cos B}$ (D) $\frac{\sin C}{\cos B}$

TR0007

8. The expression $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta}$ is equals -

(A) $\tan\theta$ (B) $\tan 2\theta$ (C) $\sin 2\theta$ (D) $\cos 2\theta$

TR0008

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)\TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

9. $\frac{1+\sin 2\theta+\cos 2\theta}{1+\sin 2\theta-\cos 2\theta}=$

(A) $\frac{1}{2}\tan\theta$ (B) $\frac{1}{2}\cot\theta$ (C) $\tan\theta$ (D) $\cot\theta$

TR0009

10. If $\tan x+\tan y=25$ and $\cot x+\cot y=30$, then the value of $\tan(x+y)$ is

(A) 150 (B) 200 (C) 250 (D) 100

TR0010

11. If $A=\tan 6^\circ\tan 42^\circ$ and $B=\cot 66^\circ\cot 78^\circ$, then -

(A) A = 2B (B) A = 1/3B (C) A = B (D) 3A = 2B

TR0011

12. In a right angled triangle the hypotenuse is $2\sqrt{2}$ times the perpendicular drawn from the opposite vertex. Then the other acute angles of the triangle are

(A) $\frac{\pi}{3}$ and $\frac{\pi}{6}$ (B) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$ (C) $\frac{\pi}{4}$ and $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$ and $\frac{3\pi}{10}$

TR0012

13. If $\tan\alpha=(1+2^{-x})^{-1}$, $\tan\beta=(1+2^{x+1})^{-1}$, then $\alpha+\beta=$

(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $\pi/2$

TR0013

14. If $3\sin\alpha=5\sin\beta$, then $\frac{\tan\frac{\alpha+\beta}{2}}{\tan\frac{\alpha-\beta}{2}}=$

(A) 1 (B) 2 (C) 3 (D) 4

TR0014

15. If $\cos\alpha=\frac{2\cos\beta-1}{2-\cos\beta}$ then $\tan\frac{\alpha}{2}\cdot\cot\frac{\beta}{2}$ has the value equal to {where $\alpha,\beta\in(0,\pi)$}

(A) 2 (B) $\sqrt{2}$ (C) 3 (D) $\sqrt{3}$

TR0015

16. If $\tan B=\frac{n\sin A\cos A}{1-n\cos^2 A}$ then $\tan(A+B)$ equals

(A) $\frac{\sin A}{(1-n)\cos A}$ (B) $\frac{(n-1)\cos A}{\sin A}$ (C) $\frac{\sin A}{(n-1)\cos A}$ (D) $\frac{\sin A}{(n+1)\cos A}$

TR0016

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Math\Module\3-CA(TR)-TE-FCG-SQT-H & D\Eng\01_Compound Angle (TR)\02_Eng.p65

17. The value of $\text{cosec}\frac{\pi}{18} - \sqrt{3}\ \sec\frac{\pi}{18}$ is a

(A) surd (B) rational which is not integral

(C) negative integer (D) natural number

TR0017

18. The value of $\cot x + \cot(60^\circ + x) + \cot(120^\circ + x)$ is equal to :

(A) $\cot 3x$ (B) $\tan 3x$ (C) $3\tan 3x$ (D) $\frac{3 - 9\tan^2 x}{3\tan x - \tan^3 x}$

TR0018

19. If $\frac{5\pi}{2} < x < 3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}$ is

(A) $-\cot\frac{x}{2}$ (B) $\cot\frac{x}{2}$ (C) $\tan\frac{x}{2}$ (D) $-\tan\frac{x}{2}$

TR0019

20. The value of $\cot 7\frac{1}{2}^\circ + \tan 67\frac{1}{2}^\circ - \cot 67\frac{1}{2}^\circ - \tan 7\frac{1}{2}^\circ$ is :

(A) a rational number (B) irrational number

(C) $2(3 + 2\sqrt{3})$ (D) $2(3 - \sqrt{3})$

TR0020

21. If $x + y = 3 - \cos 4\theta$ and $x - y = 4\sin 2\theta$ then

(A) $x^4 + y^4 = 9$ (B) $\sqrt{x} + \sqrt{y} = 16$ (C) $x^3 + y^3 = 2(x^2 + y^2)$ (D) $\sqrt{x} + \sqrt{y} = 2$

TR0021

22. If $A = \sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7}$ and $B = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7}$ then $\sqrt{A^2 + B^2}$ is equal to

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\sqrt{3}$

TR0022

23. If tan A + tan B + tan C = tan A. tan B. tan C, then -

(A) A,B,C must be angles of a triangle

(B) the sum of any two of A,B,C is equal to the third

(C) A + B + C must be n integral multiple of π

(D) None of these

TR0023

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)\TE-FOG-SOT-H & D\Eng\01_Compound Angle (TR)\02_Eng.p65

24. If $f(x) = \frac{\sin 3x}{\sin x}$, $x \neq n\pi$, then the range of values of f(x) for real values of x is -

(A) $[-1,3]$ (B) $(-\infty,-1]$ (C) $(3, +\infty)$ (D) $[-1,3)$

TR0024

25. Maximum and minimum value of $2\sin^2\theta - 3\sin\theta + 2$ is -

(A) $\frac{1}{4}, -\frac{7}{4}$ (B) $\frac{1}{4}, \frac{21}{4}$ (C) $\frac{21}{4}, -\frac{3}{4}$ (D) $7, \frac{7}{8}$

TR0025

26. For $\theta \in (0, \pi/2)$, the maximum value of $\sin\left(\theta+\frac{\pi}{6}\right)+\cos\left(\theta+\frac{\pi}{6}\right)$ is attained at $\theta =$

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

TR0026

27. Minimum value of the expression $\cos^2\theta - (6\sin\theta\cos\theta) + 3\sin^2\theta + 2$, is -

(A) $4+\sqrt{10}$ (B) $4-\sqrt{10}$ (C) 0 (D) 4

TR0027

28. The value of $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}$ is :-

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{2}$ (D) 1

TR0028

29. The exact value of $\frac{96\sin 80^\circ \sin 65^\circ \sin 35^\circ}{\sin 20^\circ + \sin 50^\circ + \sin 110^\circ}$ is equal to

(A) 12 (B) 24 (C) -12 (D) 48

TR0029

30. If m and n are positive integers satisfying

$1 + \cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta + \cos 10\theta = \frac{\cos m\theta \cdot \sin n\theta}{\sin\theta}$ then $(m + n)$ is equal to

(A) 9 (B) 10 (C) 11 (D) 12

TR0030

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

EXERCISE (O-2)

Multiple Correct Answer Type

1. If cosx = tanx, then which of the following is/are true ?

(A) $\frac{1}{\sin x}+\cos^4 x=1$ (B) $\frac{1}{\sin x}+\cos^4 x=2$ (C) $\cos^4x + \cos^2x = 1$ (D) $\cos^4x + \cos^2x = 2$

TR0031

2. If $\tan A=-\frac{1}{2}$ and $\tan B=-\frac{1}{3}$, then A + B =

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{7\pi}{4}$

TR0032

3. If $\cos(A-B)=\frac{3}{5}$ and tanA tanB = 2, then which of the following is/are correct ?

(A) cosA cosB = $-\frac{1}{5}$ (B) $\sin A\sin B=\frac{2}{5}$ (C) $\cos(A+B)=-\frac{1}{5}$ (D) $\sin A\cos B=\frac{4}{5}$

TR0033

4. If $\sin t+\cos t=\frac{1}{5}$, then $\tan\frac{t}{2}$ is equal to

(A) –1 (B) $-\frac{1}{3}$ (C) 2 (D) $-\frac{1}{6}$

TR0034

5. If $\tan^2\alpha + 2\tan\alpha.\tan2\beta = \tan^2\beta + 2\tan\beta.\tan2\alpha$, then we may have

(A) $\tan^2\alpha + 2\tan\alpha.\tan2\beta = 0$ (B) $\tan\alpha + \tan\beta = 0$

(C) $\tan^2\beta + 2\tan\beta.\tan2\alpha = 1$ (D) $\tan\alpha = \tan\beta$

TR0035

6. If $3\sin\beta = \sin(2\alpha + \beta)$, then $\tan(\alpha + \beta) - 2\tan\alpha$ is

(A) independent of α (B) independent of β

(C) dependent of both α and β (D) independent of α but dependent of β

TR0036

7. If $L = \cos^2 84^\circ + \cos^2 36^\circ + \cos 36^\circ \cos 84^\circ$

$M = \cot 73^\circ \cot 47^\circ \cot 13^\circ$

$N = 4 \sin 156^\circ \sin 84^\circ \sin 36^\circ$, then which of the following option(s) is(are) correct ?

(A) L < 1 (B) M > tan 2 (C) $N > \sin\frac{\pi}{4}$ (D) 0 < LMN

TR0037

ALLEN®

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)\TE-FOG-SOT-H & D\Eng\01_Compound Angle (TR)\02_Eng.p65

8. If $\frac{\cos 3x}{\cos x}=\frac{1}{3}$ for some angle x, $0 \le x \le \frac{\pi}{2}$, which of the following is/are true ?

(A) $\frac{\sin 3x}{\sin x}=\frac{7}{3}$ (B) $\cos 2x = \frac{2}{3}$ (C) $\tan x = \frac{1}{\sqrt{5}}$ (D) $\sin 2x = \frac{2\sqrt{5}}{6}$

TR0038

9. If $\alpha + \beta + \gamma = 2\pi$, then

(A) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2}=\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(B) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}+\tan\frac{\beta}{2}\tan\frac{\gamma}{2}+\tan\frac{\gamma}{2}\tan\frac{\alpha}{2}=1$

(C) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2}=-\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(D) $\tan\frac{\alpha}{4}\tan\frac{\beta}{4}+\tan\frac{\beta}{4}\tan\frac{\gamma}{4}+\tan\frac{\gamma}{4}\tan\frac{\alpha}{4}=1$

TR0039

10. If $x + y = z$, then $\cos^2 x + \cos^2 y + \cos^2 z - 2\cos x \cos y \cos z$ is equal to

(A) $\cos^2 z$ (B) $\sin^2 z$ (C) $\cos(x + y - z)$ (D) 1

TR0040

11. If A,B,C are angles of a triangle ABC and $\tan A \tan C = 3$; $\tan B \tan C = 6$ then, which is (are) correct?

(A) $A = \frac{\pi}{4}$ (B) $\tan A \tan B = 2$ (C) $\frac{\tan A}{\tan C}=3$ (D) $\tan B = 2\tan A$

TR0041

12. Which of the following is/are true ?

(A) $\sin\left(\frac{\pi}{4}+\theta\right)\sin\left(\frac{\pi}{4}-\theta\right)=\frac{\cos 2\theta}{2}$

(B) In a ΔABC, if $\tan A = 2$, $\tan B = 3$, then $\tan C = 1$

(C) Minimum value of $4\tan^2\theta + \cot^2\theta$ is 4 (wherever defined)

(D) $2(\sin^6 x + \cos^6 x) - 3(\sin^4 x + \cos^4 x) = -1$

TR0042

13. Let $f_n(\theta)=\sum_{r=0}^{n}\frac{1}{4^r}\sin^4\left(2^r\theta\right)$. Then which of the following alternative(s) is/are correct ?

(A) $f_2\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$ (B) $f_3\left(\frac{\pi}{8}\right)=\frac{2+\sqrt{2}}{4}$ (C) $f_4\left(\frac{3\pi}{2}\right)=1$ (D) $f_5(\pi) = 0$

TR0043

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

14. Let $y = \frac{\cos x + \cos 2x + \cos 3x + \cos 4x + \cos 5x + \cos 6x + \cos 7x}{\sin x + \sin 2x + \sin 3x + \sin 4x + \sin 5x + \sin 6x + \sin 7x}$, then which of the following hold good?

(A) The value of y when $x = \pi/8$ is not defined. (B) The value of y when $x = \pi/16$ is 1.

(C) The value of y when $x = \pi/32$ is $\sqrt{2}-1$. (D) The value of y when $x = \pi/48$ is $2+\sqrt{3}$.

TR0044

15. The value of $\frac{(\cos 11° + \sin 11°)}{(\cos 11° - \sin 11°)}$ is

(A) –tan 304° (B) tan 56° (C) cot 214° (D) cot 34°

TR0111

16. The value of $\frac{\sin x + \cos x}{\cos^3 x} =$

(A) $1 + \tan x + \tan^2 x - \tan^3 x$ (B) $1 + \tan x + \tan^2 x + \tan^3 x$

(C) $1 - \tan x + \tan^2 x + \tan^3 x$ (D) $(1 + \tan x)\sec^2 x$

TR0112

17. If $(\sec A + \tan A)(\sec B + \tan B)(\sec C + \tan C) = (\sec A - \tan A)(\sec B - \tan B)(\sec C - \tan C)$ then each side can be

(A) 1 (B) –1 (C) 0 (D) none

TR0113

18. Which of the following is correct ?

(A) $\sin 1° > \sin 1$ (B) $\sin 1° < \sin 1$ (C) $\cos 1° > \cos 1$ (D) $\cos 1° < \cos 1$

TR0114

19. If $P_n = \cos^n\theta + \sin^n\theta$ and $Q_n = \cos^n\theta - \sin^n\theta$, then which of the following is/are true.

(A) $P_n - P_{n-2} = -\sin^2\theta \cos^2\theta P_{n-4}$ (B) $Q_n - Q_{n-2} = -\sin^2\theta \cos^2\theta Q_{n-4}$

(C) $P_4 = 1 - 2\sin^2\theta \cos^2\theta$ (D) $Q_4 = \cos^2\theta - \sin^2\theta$

TR0115

20. For $0 < \theta < \pi/2$, $\tan\theta + \tan 2\theta + \tan 3\theta = 0$ if

(A) $\tan\theta = 0$ (B) $\tan 2\theta = 0$

(C) $\tan 3\theta = 0$ (D) $\tan\theta \tan 2\theta = 2$

TR0116

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)\TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question Nos. 1 to 3

Consider the polynomial $P(x) = (x - \cos 36°)(x - \cos 84°)(x - \cos 156°)$

1. The coefficient of x^2 is

(A) 0 (B) 1 (C) $-\frac{1}{2}$ (D) $\frac{\sqrt{5}-1}{2}$

TR0045

2. The coefficient of x is

(A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $-\frac{3}{4}$ (D) zero

TR0045

3. The absolute term in P(x) has the value equal to

(A) $\frac{\sqrt{5}-1}{4}$ (B) $\frac{\sqrt{5}-1}{16}$ (C) $\frac{\sqrt{5}+1}{16}$ (D) $\frac{1}{16}$

TR0045

Paragraph for Question 4 to 6

If $\cos\alpha + \cos\beta = a$ and $\sin\alpha + \sin\beta = b$ and θ is arithmetic mean between α and β, then

$\sin 2\theta + \cos 2\theta = 1 + \frac{nb(a-b)}{a^2+b^2}$ where, $n \in I$

On the basis of above information answer the following :

4. The value of n is -

(A) 0 (B) 1 (C) 2 (D) –2

TR0117

5. If for n obtained in above question, $\sin^n A = x$, then $\sin A \sin 2A \sin 3A \sin 4A$ is a polynomial in x of degree -

(A) 5 (B) 6 (C) 7 (D) 8

TR0118

6. If degree of polynomial obtained in previous question is p and $(p-5) + \sin\theta, \cos\theta, \tan\theta$ are in G.P., then $\cos^9\theta + \cos^6\theta + 3\cos^5\theta - 1 =$

(A) p (B) 0 (C) 2p (D) –1

TR0119

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

Paragraph for Question 7 and 8

Let a,b,c are respectively the sines and p,q,r are respectively the cosines of $\alpha, \alpha+\frac{2\pi}{3}$ and $\alpha+\frac{4\pi}{3}$, then

7. The value of (a + b + c) + (ab + bc + ca) is less than or equal to

(A) 0 (B) $\frac{3}{4}$ (C) 1 (D) $-\frac{3}{4}$

TR0046

8. The value of (qc – rb) is equal to $\frac{\sqrt{\alpha}}{\beta}$; α, β are coprime then -

(A) $\alpha + \beta = 5$ (B) $\alpha - \beta = -1$ (C) $\alpha - \beta = 1$ (D) $\alpha + \beta = 7$

TR0046

Matrix Match Type

9.

	List-I		**List-II**
(I)	The value of tan 20° . tan 40° . tan 60° . tan 80° is	(P)	Odd natural Number

TR0054

(II)	The value of $4\left(\sin^4\frac{\pi}{16}+\sin^4\frac{3\pi}{16}+\sin^4\frac{5\pi}{16}+\sin^4\frac{7\pi}{16}\right)$ is	(Q)	Even natural Number

TR0055

(III)	$\tan\alpha + 2\tan2\alpha + 4\tan4\alpha + 8\cot8\alpha = \lambda\cot\lambda\alpha$ then λ is	(R)	3
(IV)	$(4\cos^2 9° - 3)(4\cos^2 27° - 3) = \tan k°$, then k is	(S)	6

TR0057

(T) 1

(U) 9

Which of the following is **CORRECT** ?

(A) (I) → (P,R) ; (II) → (Q,S) ; (III) → (P,T) ; (IV) → (P, U)

(B) (I) → (P,T) ; (II) → (Q,R) ; (III) → (P,U) ; (IV) → (R, U)

(C) (I) → (R,S) ; (II) → (T,U) ; (III) → (P,T) ; (IV) → (P, Q)

(D) (I) → (P,R) ; (II) → (P,U) ; (III) → (Q,S) ; (IV) → (S, T)

module06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Mustfix\Module\3-CA(TR)-TE-FCO-SOT-H & D\Eng\01_Compound Angle (TR)\02_Eng.pmd5

10. In the following matrix match **Column-I** has some quantities and **Column-II** has some comments or other quantities

Match the each element in **Column-I** with corresponding element(s) in **Column-II**

	Column-I		**Column-II**
(A)	The value of $\left\|4\left(2\cos^3\frac{\pi}{7}-\cos^2\frac{\pi}{7}-\cos\frac{\pi}{7}\right)\right\|$ is	(P)	4
(B)	If A + B + C = π and cosA = cosB. cosC then tanB. tanC has the value equal to	(Q)	8
(C)	$4\left(\frac{\cos 20^\circ+8\sin 70^\circ\sin 50^\circ\sin 10^\circ}{\sin^2 80^\circ}\right)$ is equal to	(R)	2
(D)	The maximum value of $12\sin\theta - 9\sin^2\theta$ is	(S)	1
		(T)	6

TR0052

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FCG-SQT-H & D\Eng\01_Compound Angle (TR)\02_Eng.p65

EXERCISE (O-4)

Numerical Grid Type

1. If acute angle A = 3B and $\sin A = \frac{4}{5}$ then $\left|\frac{3\sec B - 4\operatorname{cosec} B}{2}\right|$ is

TR0047

2. If the value of $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \cos\frac{7\pi}{7} = -\frac{\ell}{2}$. Find the value of ℓ.

TR0048

3. If $\cot(\theta - \alpha)$, $3\cot\theta$, $\cot(\theta + \alpha)$ are in AP (where, $\theta \neq \frac{n\pi}{2}, \alpha \neq k\pi, n, k \in I$), then $\frac{2\sin^2\theta}{\sin^2\alpha}$ is equal to

TR0049

4. If $k_1 = \tan 27\theta - \tan\theta$ and $k_2 = \frac{\sin\theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta}$ then $\left(\frac{k_1}{k_2}\right)$ is equal to

TR0050

5. The value of the expression $\frac{1 - 4\sin 10^\circ \sin 70^\circ}{2\sin 10^\circ}$ is

TR0051

6. If $\cos(\alpha + \beta) = \frac{4}{5}$; $\sin(\alpha - \beta) = \frac{5}{13}$ & α, β lie between 0 & $\frac{\pi}{4}$, then find the value of $\tan 2\alpha$.

TR0059

7. If the value of the expression $\sin 25^\circ . \sin 35^\circ . \sin 85^\circ$ can be expressed as $\frac{\sqrt{a} + \sqrt{b}}{c}$, where a, b, c ∈ N and are in their lowest form, find the value of $\left(\frac{a + b + c}{18}\right)$.

TR0060

8. Given that $(1 + \tan 1^\circ)(1 + \tan 2^\circ)....(1 + \tan 45^\circ) = 2^n$, find $\left(\frac{n}{4}\right)$.

TR0067

9. The value of $\frac{1}{100}(\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ)$.

TR0074

10. Calculate : $\frac{2\cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$

TR0064

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)\TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

EXERCISE (JM)

1. In a ΔPQR, if $3\sin P + 4\cos Q = 6$ and $4\sin Q + 3\cos P = 1$, then the angle R is equal to :

[AIEEE-2012]

(1) $\frac{3\pi}{4}$ (2) $\frac{5\pi}{6}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{4}$

TR0083

2. The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ can be written as **[JEE-MAIN 2013]**

(1) sinA cosA + 1 (2) secA cosecA + 1 (3) tanA + cotA (4) secA + cosecA

TR0084

3. Let $f_K(x) = \frac{1}{k}\left(\sin^k x+\cos^k x\right)$ where $x \in R$ and $k \geq 1$. Then $f_4(x) - f_6(x)$ equals :

[JEE-MAIN 2014]

(1) $\frac{1}{6}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) $\frac{1}{12}$

TR0085

4. If $5(\tan^2 x - \cos^2 x) = 2\cos 2x + 9$, then the value of cos4x is :- **[JEE-MAIN 2017]**

(1) $-\frac{7}{9}$ (2) $-\frac{3}{5}$ (3) $\frac{1}{3}$ (4) $\frac{2}{9}$

TR0086

5. For any $\theta \in \left(\frac{\pi}{4},\frac{\pi}{2}\right)$, the expression $3(\sin\theta - \cos\theta)^4 + 6(\sin\theta + \cos\theta)^2 + 4\sin^6\theta$ equals :

[JEE-MAIN 2019]

(1) $13 - 4\cos^6\theta$ (2) $13 - 4\cos^4\theta + 2\sin^2\theta\cos^2\theta$

(3) $13 - 4\cos^2\theta + 6\cos^4\theta$ (4) $13 - 4\cos^2\theta + 6\sin^2\theta\cos^2\theta$

TR0087

6. The value of $\cos\frac{\pi}{2^2}\cdot\cos\frac{\pi}{2^3}\cdot.....\cdot\cos\frac{\pi}{2^{10}}\cdot\sin\frac{\pi}{2^{10}}$ is : **[JEE-MAIN 2019]**

(1) $\frac{1}{256}$ (2) $\frac{1}{2}$ (3) $\frac{1}{512}$ (4) $\frac{1}{1024}$

TR0088

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

7. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ for k = 1, 2, 3, ... Then for all $x \in R$, the value of $f_4(x) - f_6(x)$ is equal to :- **[JEE-MAIN 2019]**

(1) $\frac{5}{12}$ (2) $\frac{-1}{12}$ (3) $\frac{1}{4}$ (4) $\frac{1}{12}$

TR0089

8. The maximum value of $3\cos\theta+5\sin\left(\theta-\frac{\pi}{6}\right)$ for any real value of θ is : **[JEE-MAIN 2019]**

(1) $\sqrt{19}$ (2) $\frac{\sqrt{79}}{2}$ (3) $\sqrt{31}$ (4) $\sqrt{34}$

TR0090

9. Let α and β be two real roots of the equation $(k + 1)\tan^2 x - \sqrt{2}\lambda\tan x = (1 - k)$, where $k(\neq -1)$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is ; **[JEE-MAIN 2020]**

(1) 5 (2) 10 (3) $5\sqrt{2}$ (4) $10\sqrt{2}$

TR0091

10. The value of $\cos^3\left(\frac{\pi}{8}\right)\cdot\cos\left(\frac{3\pi}{8}\right)+\sin^3\left(\frac{\pi}{8}\right)\cdot\sin\left(\frac{3\pi}{8}\right)$ is : **[JEE-MAIN 2020]**

(1) $\frac{1}{4}$ (2) $\frac{1}{\sqrt{2}}$ (3) $\frac{1}{2\sqrt{2}}$ (4) $\frac{1}{2}$

TR0092

11. If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of $27\sec^6\alpha + 8\operatorname{cosec}^6\alpha$ is equal to :

(1) 350 (2) 500 (3) 400 (4) 250

[JEE-MAIN 2021]

TR0120

12. $16\sin(20°)\sin(40°)\sin(80°)$ is equal to :

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) 3 (D) $4\sqrt{3}$

[JEE-MAIN 2022]

TR0121

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)\TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

EXERCISE (JA)

1. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then -

[JEE 06, 3M,–1M]

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$ (C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

TR0093

2. **(a)** If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then **[JEE 2009, 4 + 4]**

(A) $\tan^2 x = \frac{2}{3}$ (B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$ (D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

TR0094

(b) For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^{6} \text{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right)\text{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is (are) -

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

TR0095

3. **(a)** The maximum value of the expression $\frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta}$ is

TR0096

(b) Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3}+1$ apart. If the chords subtend at the center, angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$ where $k > 0$, then the value of [k] is -

[**Note :** [k] denotes the largest integer less than or equal to k] **[JEE 2010, 3+3]**

TR0097

4. Let $P = \left\{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\right\}$ and $Q = \left\{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\right\}$ be two sets. Then

(A) $P \subset Q$ and $Q - P \neq \varnothing$ (B) $Q \not\subset P$

(C) $P \not\subset Q$ (D) P = Q **[JEE 2011,3]**

TR0098

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

5. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$ is equal to **[JEE(Advanced)-2016, 3(–1)]**

(A) $3-\sqrt{3}$ (B) $2(3-\sqrt{3})$ (C) $2(\sqrt{3}-1)$ (D) $2(2+\sqrt{3})$

TR0099

6. Let α and β be nonzero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. Then which of the following is/are true ? (where $\alpha,\beta \neq (2k+1)\pi$) **[JEE(Advanced)-2017, 4]**

(A) $\tan\left(\frac{\alpha}{2}\right)-\sqrt{3}\tan\left(\frac{\beta}{2}\right)=0$ (B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right)-\tan\left(\frac{\beta}{2}\right)=0$

(C) $\tan\left(\frac{\alpha}{2}\right)+\sqrt{3}\tan\left(\frac{\beta}{2}\right)=0$ (D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right)+\tan\left(\frac{\beta}{2}\right)=0$

TR0100

7. Let α and β be real numbers such that $-\frac{\pi}{4}<\beta<0<\alpha<\frac{\pi}{4}$. If $\sin(\alpha+\beta)=\frac{1}{3}$ and $\cos(\alpha-\beta)=\frac{2}{3}$, then the greatest integer less than or equal to

$$\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\cos\alpha}{\sin\beta}+\frac{\sin\beta}{\cos\alpha}\right)^2$$

is _____. **[JEE(Advanced)-2022]**

TR0122

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)\TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

ANSWER KEY

Do yourself-1

1. 10π cm **2.** $36 + 6\pi$ **3.** (a) $\left(\frac{3061}{21600}\pi\right)$ (b) $\left(\frac{257}{4320}\pi\right)$ (c) $\frac{5587}{108000}\pi$

4. $\frac{22\pi}{3}+8\sqrt{3}$ **5.** 68° 43' 38"

Do yourself-2

1. $\frac{3}{5},\frac{4}{5},\frac{5}{3}$ **2.** 2 **23.** $\frac{2xy}{x^2+y^2},\frac{2xy}{x^2-y^2}$ **27.** $\left(\frac{1}{\tan^4 A}-\tan^4 A\right)$

Do yourself-3

1. $\left\{\frac{11}{60},\frac{60}{61},\frac{61}{60}\right\};\left\{\frac{-11}{60},\frac{-60}{61},\frac{-61}{60}\right\}$ **2.** $\left\{\frac{3}{5},\frac{4}{3}\right\};\left\{\frac{-3}{5},\frac{-4}{3}\right\}$ **3.** $\left\{\frac{3}{4}\right\}$

4. $\left\{\frac{8}{17},\frac{17}{8}\right\};\left\{\frac{-8}{17},\frac{-17}{8}\right\}$ **5.** $1,\frac{3}{5}$ **6.** $\left\{\frac{3}{5},\frac{5}{13}\right\}$ **7.** $\frac{5}{13}$

8. $\left\{\frac{12}{13}\right\}$ **9.** $\left\{\pm1,\pm\frac{1}{\sqrt{3}}\right\}$ **10.** $\left\{\frac{1}{2},-\frac{1}{3}\right\}$ **11.** $\pm\frac{1}{\sqrt{2}}$

12. $\left\{1\pm\sqrt{2}\right\}$ **13.** $\pm\frac{2x(x+1)}{2x^2+2x+1};\pm\frac{2x+1}{2x^2+2x+1}$

Do yourself-4

2. (a) $\frac{3}{2}$ (b) $4\frac{1}{3}$ (c) $\frac{1+\sqrt{3}}{4}$ (d) $-\frac{\sqrt{3}-1}{2\sqrt{2}}$

Do yourself-5

1. **(i)** 8 **2.** (a) 1 (b) 0

3. (a) $-\left(\frac{\sqrt{3}+1}{2}\right),-\frac{4}{\sqrt{3}}$ (b) $\left(\sqrt{2},-2\right)$ (c) $\frac{\sqrt{3}+1}{2},-\frac{4}{\sqrt{3}}$ (d) 0,2

4. (a) –cos(25°) (b) –cos(28°) (c) cot 25° (d) cot(26°)

5. (a) negative (b) positive (c) zero

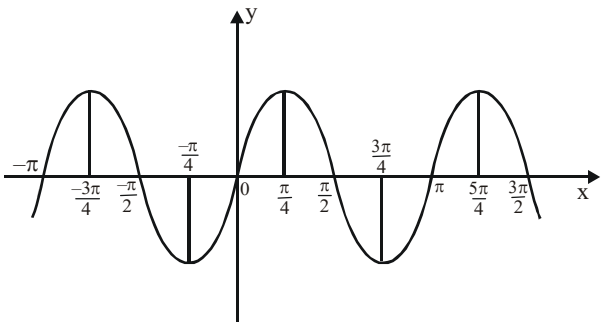
6. (a) positive (b) positive (c) negative **7.** $-\frac{1}{\sqrt{3}}$ and $\frac{\sqrt{2}}{\sqrt{3}}$; $\frac{1}{\sqrt{3}}$ and $-\frac{\sqrt{2}}{\sqrt{3}}$

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TRI)-TE-FOG-SOT-H&D\Eng\01_Compound Angle (TRI)\02_Eng.p65

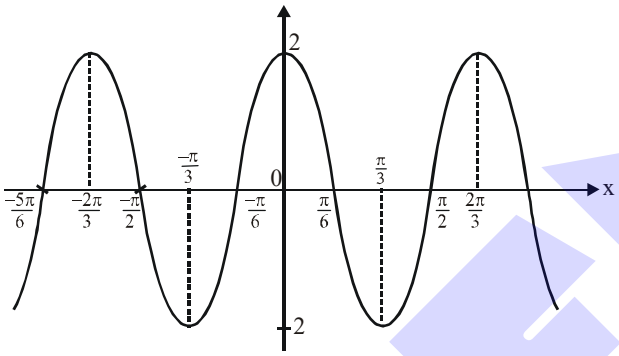
Do yourself-6

3. B

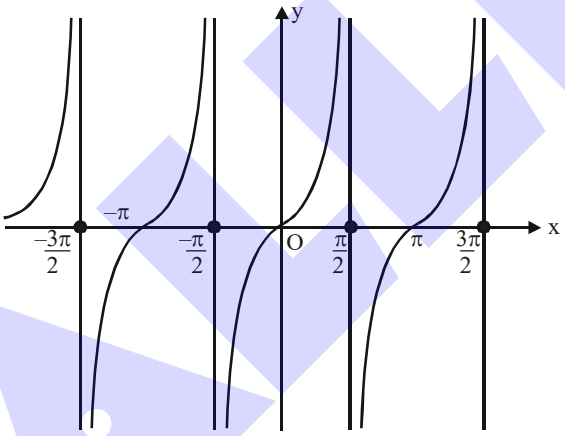
4. (a)



(b)



(c)



5. 2

Do yourself-7

1. (a) $\frac{187}{205}$ (b) $\frac{-133}{205}$ (c) $\frac{-84}{205}$ (d) $\frac{156}{205}$

3. $\sin(\alpha-\beta)=-\frac{133}{205}\text{ or }\frac{187}{205}$; $\cos(\alpha+\beta)=-\frac{84}{205}\text{ or }\frac{156}{205}$

4. $\sin(\alpha+\beta)=\frac{220}{221}\text{or}\frac{140}{221}$; $\cos(\alpha-\beta)=\pm\frac{171}{221}\text{or }\pm\frac{21}{221}$; $\sin(\alpha+\beta)=\frac{220}{221}\text{or}\frac{140}{221}$; $\tan(\alpha+\beta)=\pm\frac{220}{21}\text{or }\pm\frac{140}{171}$

13. 3 and $\frac{9}{13}$

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Maths \Module\3-CA(TR)\TE-FOG-SOT-H & D\Eng\01_Compound Angle (TR)\02_Eng.p65

Do yourself-8

1. $\frac{1}{\sqrt{3}}$ **2.** $\cos 2\theta - \cos 12\theta$ **3.** $\cos 12^\circ - \cos 120^\circ$

Do yourself-11

8. D **9.** A

Do yourself-12

1 (a) $\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$ (b) $\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$ (c) $\sqrt{2}-1$ **4.** $\frac{\pm 2\sqrt{2} \pm \sqrt{3}}{6}; \frac{\pm 7\sqrt{3} \pm 4\sqrt{2}}{18}$

5. $\frac{16}{305}; \frac{49}{305}$ **6.** $\frac{7}{5\sqrt{2}}$ **7.** $\pm\frac{1}{3}; \pm\frac{3}{4}$

8. (a) $\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}}$; (b) $\frac{\sqrt{4+\sqrt{2}+\sqrt{6}}}{2\sqrt{2}}$; (c) $\sqrt{4+2\sqrt{2}} - \left(\sqrt{2}+1\right)$ **9.** $\pm\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$

Do yourself-13

1. (a) $-\frac{1}{2}$ (b) $\frac{\sqrt{5}+1}{8}$ **2.** $\frac{\sqrt{5}-1}{8}$ **3.** $\frac{5}{16}$ **4.** 1 **5.** $\frac{1}{2^7}$

6. 1 **9.** 4 **10. B**

Do yourself-14

1. 0 **2.** 1

Do yourself-15

1. 7 & –7 **2.** $-\frac{9}{8}$ **3.** $4+\sqrt{10}$ & $4-\sqrt{10}$ **4.** (i) 2, – 1 (ii) 3,0 **5.** $\frac{1}{2}$

6. $\left\{\frac{13}{3}, -1\right\}$ **7.** 25 **8.** {25,–25} **9.** $\left\{\sqrt{13}, -\sqrt{13}\right\}$ **10.** (i) 30 (ii) 18

Do yourself-16

1. 0 **3.** $\frac{\cot 1^\circ}{90}$ **5.** (a) $\frac{1}{2}$ (b) 0 **6.** C

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H&D\Eng\01_Compound Angle (TR)\02_Eng.p65

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	B	A	B	C	B	B	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	B	D	D	A	D	D	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	D	D	A	B	B	B	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C	B,D	B,C	B,C	B,C,D	A,B	A,B,C,D	A,B,C,D	A,D	C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,D	A,B,C,D	C,D	B,D	A,B,C,D	B,D	A,B	B,C	A,B,C,D	C,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A	C	B	C	A	B	A,B,C,D	A,C	A	
Q.10	A	B	C	D						
	S	R	Q	P						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5	3	3	2	1	1.69 or 1.70	1.33	5.75	0.04	1.73

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	4	1	1	3	4	1	2	3
Que.	11	12								
Ans.	4	B								

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	
Ans.	B	(a) A,B; (b) C,D	(a) 2; (b) k = 3	D	C	Bonus	1	

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Mutiki Module\3-CA(TR)-TE-FCO-SOT-H & D\Eng\01_Compound Angle (TR)\02_Eng.pmd 5